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Revealing Electrochemically Generated Local High Order Polyiodides (I_{2n+1}^-, n = 1–3) on Pt Ultramicroelectrode

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Zn-polyiodide redox flow batteries (RFB) with a highly concentrated ZnI_2 electrolyte have been considered as promising grid scale energy storage devices. Despite a continuous energy density enhancement in Zn-polyiodide RFBs after their first introduction as a prototype by Li et al. in 2015, a fundamental understanding of electrochemical/chemical reactions occurring in Zn-polyiodide RFBs still has not been well known. One of the first subjects to understand with regard to the battery system would be chemical speciation analyses, which would dynamically change depending on the concentration of Zn^{2+} and I^- in an electrolyte solution during charging/discharging. In this article, we present the speciation of electrochemically generated polyiodides (I_{2n+1}^-) locally existing on a Pt ultramicroelectrode (UME) during electro-oxidation of I^- in an acidic aqueous solution with highly concentrated ZnI_2 via voltammetric titration. When I^- is electrochemically oxidized, an iodine film immediately forms on a Pt UME, which is designated as I_2-F. After the formation of I_2-F on a Pt UME, I^- would transfer from the aqueous to the I_2-F phase and electro-oxidized to form I_{2n+1}^- at equilibrium based on the model suggested by Gileadi et al. From the mechanistic model, we could estimate the concentration of I^- in I_2-F, $C_{I^-(I_2-F)}$ as a function of the bulk concentration of free I^- in ZnI_2 solutions, $C_{I^-(aq),free}$ by a measurement of the steady state current, i_{ss} associated with the electro-oxidation of I^-. Then, $-\log C_{I^-(I_2-F)}$ can be plotted as a function of $C_{I^-(aq),free}$. The graph shows several plateaus, which are indicative of I^--buffer regions resulting from I_{2n+1}^--dissociation reactions to I^- and nI_2 with corresponding equilibrium constants, $K_{eq,n}$ in I_2-F. Eventually, the fractional distribution of $I_{2n+1}^-(I_2\text{-}F)$ as a function of $C_{I_2(I_2-F)}$ can be obtained. We performed the presented electrochemical analysis within an acidic ZnI_2 solution with $C_{I^-(aq),free} < 4$ M and found that I_3^-, I_5^-, and I_7^- would form in I_2-F, among which I_7^- would be the main I_{2n+1}^-. Nonetheless, we found I_3^- to be the most stable polyiodide species in an aqueous solution from the computational calculations, indicating that both I_5^- and I_7^- only exist locally in I_2-F during the electro-oxidation of I^- on a Pt UME. Lastly, we estimated the fraction of I_{2n+1}^- in I_2-F as a function of the overall rate constant of $I_3^-(aq)$-formation in various concentrations of free I^- in aqueous ZnI_2 solutions.
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Aqueous Zn-polyiodide redox flow batteries (RFBs) have led to significant impacts with regard to various studies related to RFBs due to their high-energy density, which is comparable to that of Li-ion batteries.[1–6] The main reason for the high energy density of Zn-polyiodide RFBs could be attributed to the extremely high solubility of ZnI_2 in an aqueous solution: ~7.0 M in water.[1,4] Secondly, the solubility of electrochemically generated solid I_2 is greatly enhanced within an aqueous solution as the concentration of I^- increases due to the formation of I_3^-, enabling the continuous and stable electrolysis of I^- during the charging of Zn-polyiodide RFBs.

Despite their great potential as a grid-scale energy storage system, the detailed physicochemical characteristics of the two half-redox reactions within highly concentrated ZnI_2 aqueous media have been largely unknown; charging of the battery can be described as follows:

$$\text{On cathode:}\quad 3I^- \rightarrow I_3^- + 2e^-$$

$$\text{On anode:}\quad Zn^{2+} + 2e^- \rightarrow Zn$$

Since the capacity of Zn-polyiodide RFBs is primarily governed by the I_3^-/I^- half redox reaction,[1] the I_3^-/I^- redox reaction mechanism should be deeply understood to further improve Zn-polyiodide RFBs.

Chemical speciation on an electrode during the I_3^-/I^- half-redox reaction should be well characterized as a first step toward elucidating the corresponding reaction pathway during the electro-oxidation of I^- to I_3^- and vice versa. From this perspective, however, the speciation of polyiodides, I_{2n+1}^- ($n > 1$), possibly generated by the electro-oxidation of I^- on an electrode within highly concentrated ZnI_2 media is barely understood. It has been reported that I_3^- is the main I_{2n+1}^- within an aqueous solution due to the instability of higher orders of polyiodides such as I_5^- and I_7^-.[7,8] On the other hand, Li et al. suggested the possible formation of I_5^- and I_7^- during the charging of Zn-polyiodide RFBs and gave qualitative evidence of I_5^- based on Raman spectroscopic analyses of solutions after charging RFBs.[1] Weaver et al. also reported evidence of I_5^- during the electro-oxidation of I^- on an Au electrode through surface-enhanced Raman spectroscopy (SERS).[9] Recently, Sakai et al. reported a stable hydrophobic polyiodide complex including I_3^- and I_5^- during the cathodic oxidation of iodide in the presence of propylene carbonate in an aqueous solution.[10]

Previously, our group reported the new electro-analytical approach toward quantitatively analyzing the speciation of quaternary ammonium polybromides (QBr_{2n+1}) via voltammetric Br_3^--titration on a Pt ultramicroelectrode (UME) and showed the fractional diagram of each Br_{2n+1}^-, where n is equal to 1 up to 5 within existing QBr_{2n+1} ionic liquids. The introduced voltammetric titration approach was possible because a steady state current, i_{ss} was only associated with the concentration of "free" Br_3^-, which was equilibrated with Br_{2n+1}^- and $(n-1)Br_2$ in QBr_{2n+1} ionic liquids.[11]

In this article, we expand our electrochemical approach to quantitatively analyze I_{2n+1}^--speciation existing on an electrode surface during the steady state electro-oxidation of I^- through the voltammetric titration of I_2 films by I^- on a Pt UME. From experimental and computational calculation results, we found that I_7^- would be the main I_{2n+1}^- locally existing in the vicinity of Pt UME during the electro-oxidation of I^-, while I_3^- was the primary polyiodide in a bulk aqueous solution.

Experimental

Chemicals.—All solutions were prepared with deionized Milli-Q water. Chemicals were purchased and used as-is without further purification as follows: zinc iodide (ZnI_2 > 98%) from Sigma-Aldrich and sodium iodide (NaI > 98.5%), anhydrous sodium

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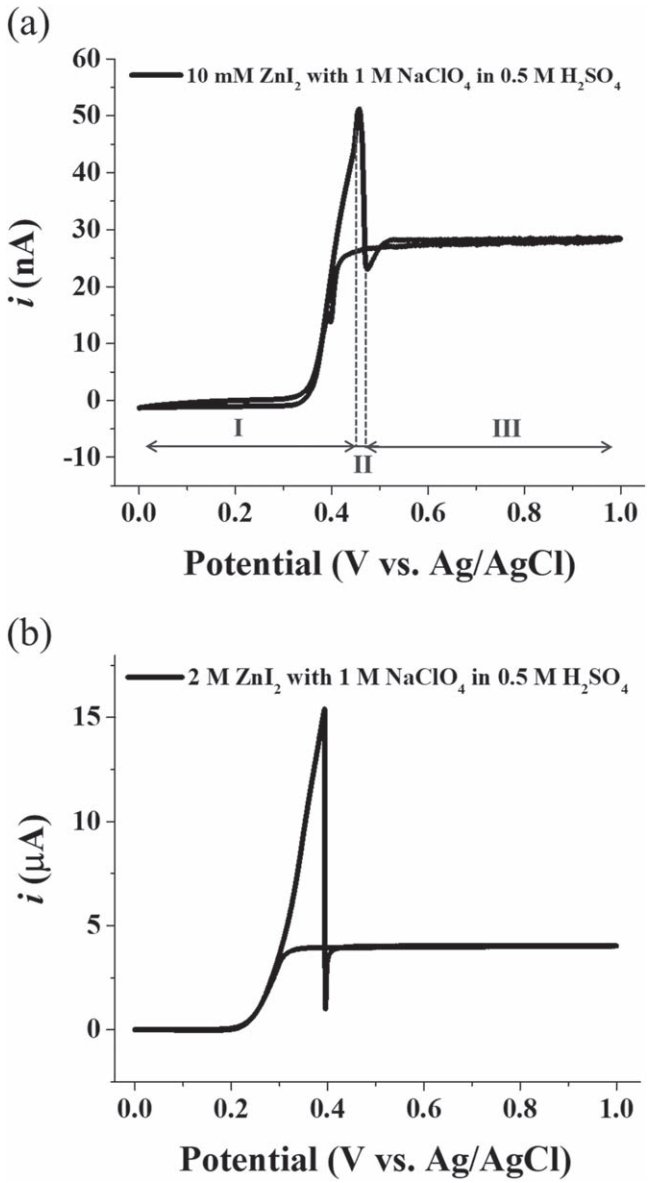

Figure 1. CV data measured in a 0.5 M H_2SO_4 aqueous solution containing (a) 10 mM and (b) 2 M ZnI_2 with 1 M $NaClO_4$.

perchlorate ($NaClO_4$ > 98.5%) from Daejung Chemical (Cheongwon, South Korea).

Instruments and measurements.—A CHI-600e potentiostat (CH Instruments, Austin, TX) was used for all electrochemical measurements. Three electrodes were used in the electrochemical cell; either a Pt ultramicroelectrode (UME) with a radius of $a = 5$ μm or a Pt macro disk electrode with a radius of $a = 1$ mm served as the working electrode, Ag/AgCl (1 M KCl) served as the reference electrode, and Pt wire served as the counter electrode. All electrodes were purchased from CH Instruments. Electrochemical measurements were performed within a home-built Faraday cage at room temperature under ambient conditions. All aqueous solutions were de-aerated with Ar for 15 min. The iodide solutions in all the electrochemical measurements included 0.5 M H_2SO_4 and 1 M $NaClO_4$. The reason for the addition of excess H_2SO_4 in this study was to avoid the potential precipitation of either ZnO or $Zn(OH)_2$; Figure S1 (available online at stacks.iop.org/JES/167/046509/mmedia) as an example shows a photo of a vial containing 3 M ZnI_2 and 1 M $NaClO_4$ without any acid.

COMSOL Multiphysics v.5.2a (COMSOL, Inc., Burlington, MA) was used for the finite element analysis. Cyclic voltammogram (CV) simulations were performed using DigiElch-Professional v6.F (ElchSoft.com).

Results and Discussion

Cyclic voltammetry for electro-oxidation of I^- in an acidic aqueous solution containing ZnI_2.—Figure 1a shows a cyclic voltammogram (CV) on a Pt ultramicroelectrode (UME) measured in a 0.5 M H_2SO_4 acidic aqueous solution containing 10 mM ZnI_2 with 1 M $NaClO_4$ as the supporting electrolyte. The CV was same as those reported in previous articles.[12–14] The forward voltammogram could be divided into three regions. At the low positive potential region (Region-I in Fig. 1a), the current exponentially increased as a function of the biased potentials and became 51.2 nA at 0.46 V. Then, the current abruptly decreased to 23.1 nA until the potential was biased to 0.48 V (Region-II). The current, however, did not drop to 0, but partially recovered and reached a steady state with the current, $i_{ss} = 28.4$ nA, designated as Region-III. This voltammetric feature became clearer as the concentration of ZnI_2, $C_{ZnI_2(aq)}$ increased to a few molar concentration levels as shown in Fig. 1b, where the CV data is measured in a 0.5 M H_2SO_4 aqueous solution with $C_{ZnI_2(aq)} = 2$ M.

In order to identify the electrochemical reaction for the oxidation current in Region-I, the diffusion coefficient of $I^-(aq)$, $D_{I^-(aq)}$ was electrochemically estimated through voltammetric analysis on a Pt macro-disk electrode in an acidic solution with $C_{ZnI_2(aq)} = 1$ mM as shown in Fig. S2a, which showed redox-paired peaks during forward and reverse scans without the complicated voltammetric features observed from the CV on a Pt UME in Fig. 1. The anodic and cathodic peak current (i_{pc} and i_{pa}) from the CV on a Pt macro-disk electrode were both linearly proportional to the square root of the scan rate, $\upsilon^{1/2}$ as shown in Fig. S2b.

It has been well known that $I^-(aq)$ is electrochemically oxidized to $I_2(aq)$ in an aqueous solution with a low concentration of I^-;[15] therefore, we considered that the CV in Fig. S2a could be attributed to the $I_2(aq)/I^-(aq)$ redox reaction and both $D_{I^-(aq)}$ and $D_{I_2(aq)}$ were estimated as 1.47×10^{-5} and 1.72×10^{-5} $cm^2\,s^{-1}$, respectively, from the plots in Fig. S2b, which were also in good agreement with those reported previously.[13,15,16]

The CV measured in an acidic solution with $C_{ZnI_2(aq)} = 1$ mM on a Pt UME can be seen in Fig. S3; $D_{I^-(aq)}$ estimated from i_{ss} was in good agreement with the value from the CV on a Pt macro disk electrode. i_{ss} on a UME could be defined with the following equation:

$$i_{ss} = 4n_{e^-}FC_{I^-(aq)}D_{I^-(aq)}a \qquad [1]$$

where, n_{e^-} is the electron number of 1 per $I^-(aq)$, F is the Faraday constant (96485 C mol^{-1}), $C_{I^-(aq)}$ is the bulk concentration of I^- in an aqueous solution, and a is the radius of a Pt UME, which is 5 μm.

Next, we estimated the theoretical i_{ss} associated with the electro-oxidation of I^- to I_2 on a Pt UME in the solution with $C_{ZnI_2(aq)} = 10$ mM; i_{ss} was close to the maximum current at Region-I of the corresponding CV as can be seen in Fig. S4, which would indicate that the forward voltammogram in Region-I of the CV could be attributed to the electro-oxidation of I^- to I_2. Additionally, we fit the digital simulation result based on the suggested reaction to the experimental voltammetric curve in Region-I of the CV (Fig. S4); the suggested elementary reactions and corresponding reaction parameters are listed in Table SI. From the simulation results, we concluded that I^- in an aqueous phase was electro-oxidized to I_2 under mass-transfer conditions on a Pt UME at Region-I of the CV in Fig. 1.

With regard to the electro-oxidation of I^- to I_2 in Region I of the CV (Fig. 1a), the phase of the produced I_2 needs to be determined as either $I_2(aq)$ or $I_2(s)$. At first, we would follow the hypothesis that the produced I_2 in Region-I is a soluble species, i.e. $I_2(aq)$. In this

case, we consider the following reactions;

$$I_2(aq) + 2e^- \rightleftharpoons 2I^-(aq) \quad [2]$$

$$I^-(aq) + I_2(aq) \rightleftharpoons I_3^- \quad K_{stab,I_3^- \ from \ I_2(aq)} \quad [3]$$

Where, $K_{stab,I_3^- \ from \ I_2(aq)}$ is the stability constant of $I_3^-(aq)$ in the reaction 3, which is determined by the following equation;

$$\ln K_{stab,I_3^- \ from \ I_2(aq)} = \frac{2F}{RT}(E^{o'}{}_{I_2(aq)/I^-(aq)} - E^{o'}{}_{I_3^-(aq)/I^-(aq)}) \quad [4]$$

Where, R is the gas constant, T is the temperature, 25 °C, and $E^{o\prime}$ is the formal potential of each redox reaction. If we would assume that the both $I_3^-(aq)/I^-(aq)$ and $I_2(aq)/I_3^-(aq)$ redox reactions are electrochemically reversible, both the $E^{o\prime}$ of each redox reaction could be considered as a half-wave potential $E_{1/2}$ of the corresponding voltammetric curve. Then, $E_{1/2,I_2(aq)/I^-(aq)}$ can be defined as follows;[17]

$$E_{1/2,I_2(aq)/I^-(aq)} = \frac{2E_{1/2,I_3^-(aq)/I^-(aq)} + E_{1/2,I_2(s)/I_3^-(aq)}}{3} \quad [5]$$

We eventually express Eq. 4 by the combination with Eq. 5 as follows;

$$\ln K_{stab,I_3^- \ from \ I_2(aq)} = \frac{2F}{3RT}(E_{1/2,I_2(aq)/I_3^-(aq)} - E_{1/2,I_3^-(aq)/I^-(aq)}) \quad [6]$$

Since $K_{stab,I_3^- \ from \ I_2(aq)}$ of reaction 3 was reported as 698 M^{-1},[18] the difference between the two $E_{1/2}$ should be *ca.* 0.25 V from a voltammogram in an aqueous solution with 10 mM ZnI_2. This result indicates that the two sequential voltammetric waves associated with $I_3^-(aq)/I^-(aq)$ and $I_2(aq)/I_3^-(aq)$ should be distinctly observed. However, we did not observe the two resolved voltammetric waves in Region-I for the CVs of Figs. 1a–1b in either the 10 mM or 2 M ZnI_2 solutions, which infers the first hypothesis would be not rationale.

As the alternative case, we, now, could consider the second hypothesis that the produced I_2 is solid in Region-I. The solubility product of the following reaction, K_{sp,I_2} has been known as *ca.* 1.3×10^{-3} M;[19]

$$I_2(s) \rightleftharpoons I_2(aq) \quad K_{sp,I_2} \quad [7]$$

Therefore, $K_{stab,I_3^- \ from \ I_2(s)}$ in the formation reaction of $I_3^-(aq)$ from $I_2(s)$ and $I^-(aq)$ can be estimated by product between $K_{stab,I_3^- \ from \ I_2(aq)}$ and K_{sp,I_2}, which is 0.91;

$$I_2(s) + I^-(aq) \rightleftharpoons I_3^-(aq) \quad K_{stab,I_3^- \ from \ I_2(s)} \quad [8]$$

In this case, we could also estimate the difference of $E_{1/2}$ based on Eq. 6, and find that $E_{1/2,I_3^-(aq)/I^-(aq)}$ would be 3.6 mV more positive than $E_{1/2,I_2(s)/I_3^-(aq)}$, resulting in unresolved one voltammetric curve during electro-oxidation of $I^-(aq)$. Based on the above analysis, we could conclude that $I^-(aq)$ would be electro-oxidized to mainly $I_2(s)$ in Region-I.

The current fell in Region-II. This phenomenon has been generally explained by the formation of a thick I_2 film.[20] As the local concentration of I_2 in the vicinity of a Pt UME increased, I_2 forms a solid film on a Pt electrode, which blocks the electro-oxidation of I^- on a Pt UME, resulting in a dramatic current decrease. However, the current from the electro-oxidation of I^- was partially recovered and reached a steady state in Region-III. Gileadi et al.[15] proposed a semiconductor and ionic conductivity model to explain the partial current recovery. In this model, I^- is dissolved in an I_2 film (I_2-F) and a certain amount of I^- in I_2-F is the key factor toward enhancing the conductivity of I_2-F. This mechanism suggested that iodine is not completely dissolved but partially remained in the steady state where I^-, I_2 and I_3^- (or higher polyiodides) are equilibrated.

The suggested reaction model for formation of polyiodides (I_{2n+1}^-, $n \geqslant 1$) in I_2-F at steady state.—Based on the proposed mechanism by Gileadi[15] et al., we suggested the reaction model for the formation of polyiodides (I_{2n+1}^-, $n \geqslant 1$) in I_2-F at the steady state, which is schematically described in Fig. 2. In this model, we considered that $I^-(aq)$ mass transfers through I_2-F, and i_{ss} is considered to be associated with the electro-oxidation of I^- to I_2 in I_2-F.

$$I^-(aq) \rightleftharpoons I^-(I_2 - F) \quad [9]$$

$$2I^-(I_2 - F) \rightarrow I_2(I_2 - F) + 2e^- \text{on a Pt UME} \quad [10]$$

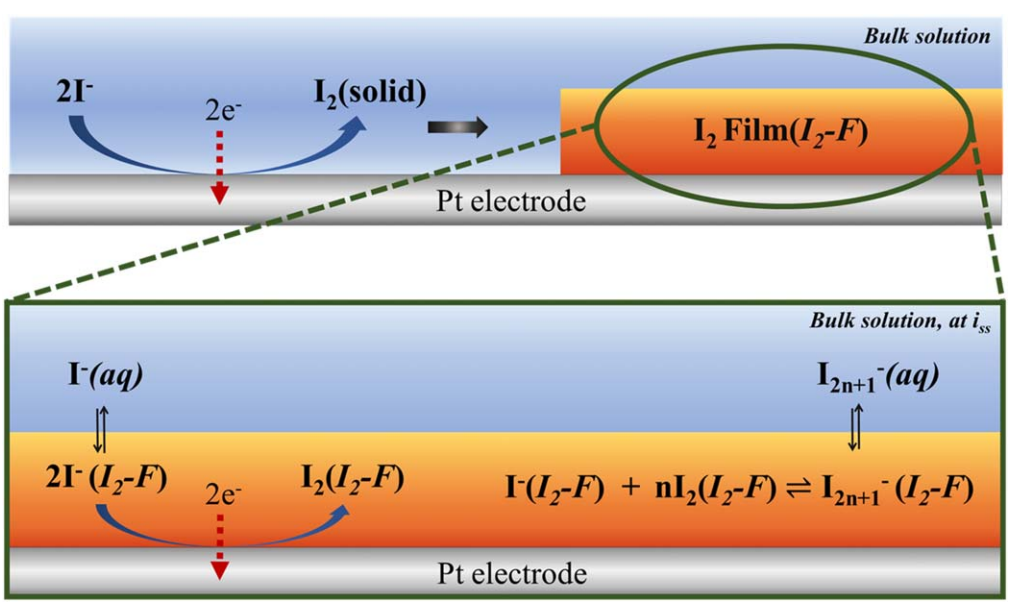

Figure 2. A schematic description for the I_{2n+1}^--formation through the electro-oxidation of I^- in the steady state under mass-transfer limited conditions followed by the proposed mechanism by Gileadi et al.

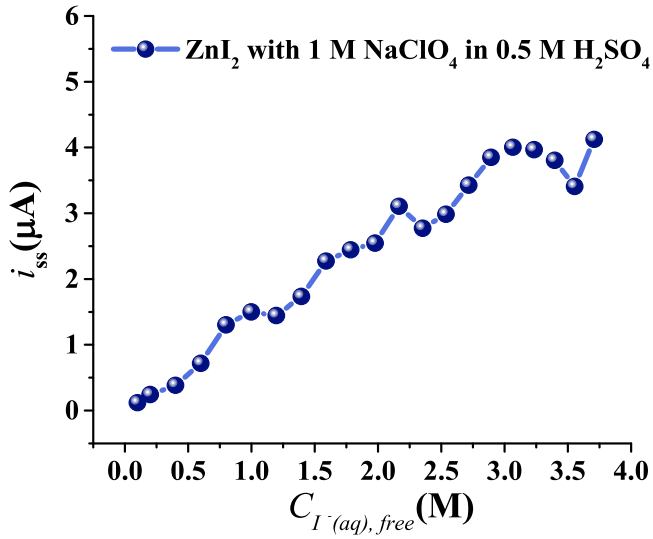

Figure 3. i_{ss} vs $C_{I^-(aq),free}$ plots estimated from CV data measured in acidic solutions with various $C_{ZnI_2(aq)}$.

In I_2-F, I^-, I_2, and I_{2n+1}^- are in equilibrium with the corresponding equilibrium constant, $K_{eq,n}$ as follows:

$$I_{2n+1}^-(I_2 - F) \rightleftharpoons I^-(I_2_F) + nI_2(I_2_F) \quad K_{eq,n} \qquad [11]$$

The equilibrium constant, $K_{eq,n}$ for $I_{2n+1}^-(I_2$-$F)$ dissociation to $I^-(I_2$-$F)$ and $nI_2(I_2$-$F)$ in the reaction 11 could be expressed as follows:

$$K_{eq,n} = \frac{\alpha_{I^-(I_2-F)}(\alpha_{I_2(I_2-F)})^n}{\alpha_{I_{2n+1}^-(I_2-F)}} \qquad [12]$$

Where $\alpha_{I_{2n+1}^-(I_2-F)}$, $\alpha_{I^-(I_2-F)}$, and $\alpha_{I_2(I_2-F)}$ are the activities of I_{2n+1}^-, I^-, and I_2 in I_2-F, respectively. When we substitute the activity coefficients of each species to a mean activity coefficient, $\gamma_\pm$, Eq. 12 can be converted to the following logarithmic equation:

$$-\log C_{I^-(I_2-F)} = -\log K'_{eq,n} + log\frac{(C_{I_2(I_2-F)})^n}{C_{I_{2n+1}^-(I_2-F)}} \qquad [13]$$

Where, $K'_{eq,n}$ is the corrected equilibrium constant defined as $K'_{eq,n} = K_{eq,n}/\gamma_\pm^n$.[11]

Eventually, the fractions of I_{2n+1}^- species in I_2-F, $f_{I_{2n+1}^-(I_2-F)}$ could be estimated using the following equation;

$$f_{I_{2n+1}^-(I_2-F)} = [I_2(I_2 - F)]^n \Big/ \left(1 + \sum_{k=1}^{n} \frac{[I_2(I_2 - F)]^n}{K'_{eq,n}}\right) \qquad [14]$$

An estimation of $K'_{eq,n}$ and $f_{I_{2n+1}^-}$ at the steady state based on the suggested mechanistic models.—Since the electro-oxidation of I^- on a Pt UME was only associated with free I^- that was uncoordinated with Zn^{2+} to form ZnI_x^{2-x} complexes, we estimated the concentration of free I^-, $C_{I^-(aq),free}$, from the corresponding ZnI_2 solutions in our previous report.[13] Fig. 3 shows i_{ss} vs $C_{I^-(aq),free}$ estimated from CV data measured in acidic solutions with $C_{I^-(aq),free} < 4$ M (see Fig. S5); the estimated i_{ss} fluctuated as $C_{I^-(aq),free}$ became equal to or greater than 4 M as shown in Fig. S6. Interestingly, i_{ss} did not linearly increase as a function of $C_{I^-(aq),free}$; the plot showed three plateau regions at different $C_{I^-(aq),free}$. Similar i_{ss} behavior was also observed in condensed quaternary ammonium polybromide (QBr_{2n+1}) ionic liquids, which were previously reported[11]; we revealed that the i_{ss}-plateaus would have primarily originated from the equilibria of polybromide dissociation in QBr_{2n+1} ionic liquids.

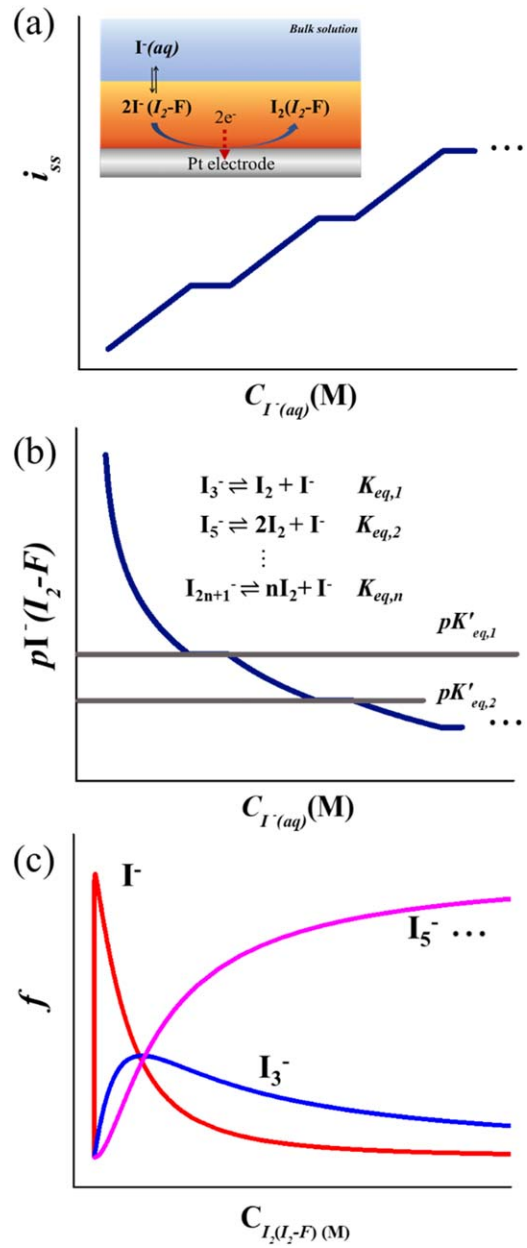

Figure 4. Representative data analyses based on the reaction model in Fig. 2 made through electrochemical analysis: (a) i_{ss} vs $C_{I^-(aq),free}$, (b) $pI^-(I_2$-$F)$ vs $C_{I^-(aq),free}$, and (c) $f_{I_{2n+1}^-(I_2-F)}$ vs $C_{I_2(I_2-F)}$ plots.

Figure 4 shows a description of the representative electrochemical data analyses based on the suggested model described in Fig. 2 for an estimation of $K'_{eq,n}$ during polyiodide dissociation and the corresponding $f_{I_{2n+1}^-}$ in I_2-F on a Pt UME. As shown in Fig. 4a, i_{ss} would be mainly proportional to the concentration of I^- in I_2-F, $C_{I^-(I_2-F)}$. $C_{I^-(I_2-F)}$ increased as a function of $C_{I^-(aq)}$ due to the equilibrium at the interface between I_2-F and an aqueous phase (reaction 9). However, $C_{I^-(I_2-F)}$ was maintained in the region where $C_{I_2(I_2-F)}/C_{I_3^-(I_2-F)}$ became ~1 as described in Eq. 13, resulting in a constant i_{ss}. The plot of i_{ss} vs $C_{I^-(aq)}$ could be converted to $pI^-(I_2$-$F)$ $(= -\log C_{I^-(I_2-F)})$ vs $C_{I^-(aq),free}$ as shown in Fig. 4b. I^--buffer regions were observed at $pI^-(I_2$-$F) \sim pK'_{eq,n}$, where n of I_{2n+1}^- dissociation sequentially increased as $pI^-(I_2$-$F)$ decreased. Eventually, $f_{I_{2n+1}^-(I_2-F)}$ could be obtained from the measured $pK'_{eq,n}$ based on Eq. 14 as shown in Fig. 4c.

With regard to the conversion of the i_{ss} plots in Fig. 3 to $C_{I^-(I_2-F)}$ based on the suggested reaction model, we tried to adopt the

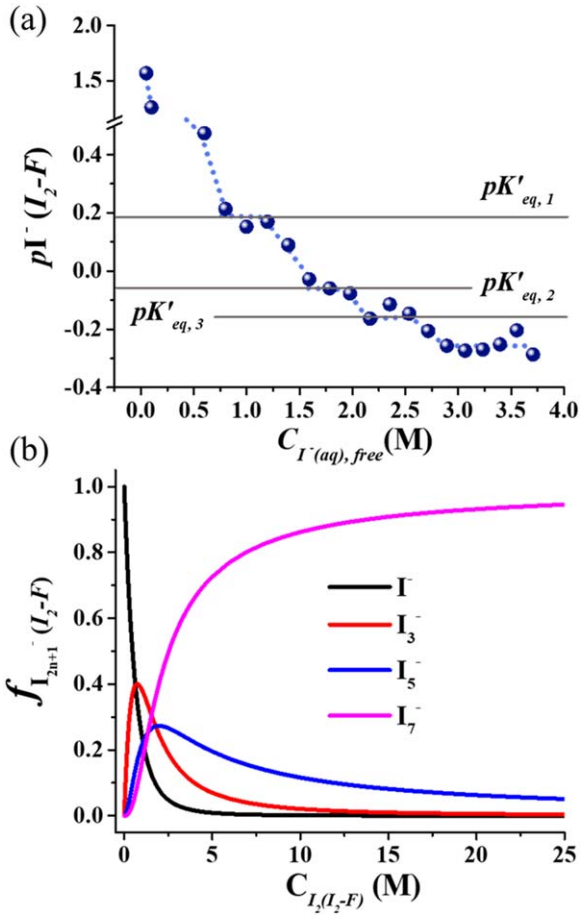

Figure 5. Experimentally obtained (a) $pI^-(I_2\text{-}F)$ vs $C_{I^-(aq),free}$ and (b) $f_{I_{2n+1}^-(I_2-F)}$ vs $C_{I_2(I_2-F)}$ based on the mechanistic model for electro-oxidation of I^- as shown in Fig. 2.

diffusion coefficient of I^- in I_2-F, $D_{I^-(I_2-F)}$ as 3×10^{-8} cm^2 s^{-1} from a previous report by Brukenstein et al.[21] Then, the estimated $C_{I^-(I_2-F)}$ was hundreds of times higher than that for the corresponding aqueous solutions shown in Fig. S7a. If we would follow this obtained result, I^- would be highly stabilized in I_2-F compared to that in the aqueous phase. Additionally, $f_{I_{2n+1}^-(I_2-F)}$ from the estimated $K'_{eq,n}$ (see Fig. S7b) also showed that I^- exists in I_2-F even at high $C_{I_2(I_2-F)}$ as shown in Fig. S7c.

This result would not be reasonable because it has been well known that I^- in I_2-F would eventually react with I_2 to form polyiodides.[1] Therefore, $C_{I^-(I_2-F)}$ should be the same or less than that in the bulk aqueous solution; i.e. the partition coefficient in reaction 9 was the same or less than 1. Iodide species in I_2-F forming within an aqueous solution with a reasonably low concentration of ZnI_2 could be considered as I^-, I_2, and I_3^- without higher orders of I_{2n+1}^-; we found voltammetric characteristics indicating the formation of I_2-F were observed in the solution with 3mM ZnI_2 as shown in Fig. S8. Under the assumption that $C_{I^-(I_2-F)}$ was the same as $C_{I^-(aq),free}$, we could estimate $D_{I^-(I_2-F)}$ at equilibrium from i_{ss} of the CV in this condition, which would be associated with the electro-oxidation of I^- to I_3^- in I_2-F: $D_{I^-(I_2-F)} = 1.1 \times 10^{-5}$ cm^2 s^{-1}. The estimated large $D_{I^-(I_2-F)}$ would support fast I^--transfer through I_2-F based on a Grotthus-type hopping mechanism of I^- in a condensed iodine phase.[22–24]

From the $D_{I^-(I_2-F)}$, $pI^-(I_2\text{-}F)$ as a function of $C_{I^-(aq),free}$ can be seen in Fig. 5a. From the $pI^-(I_2\text{-}F)$ vs $C_{I^-(aq),free}$ plot, the three I^--buffer regions were observed and $K'_{eq,1-3}$ values for the dissociation of $I_3^-(I_2\text{-}F)$, $I_5^-(I_2\text{-}F)$ and $I_7^-(I_2\text{-}F)$ were estimated from corresponding $pI^-(I_2\text{-}F)$ values at the three plateaus: $K'_{eq,1} = 0.65$, $K'_{eq,2} = 1.14$, and $K'_{eq,3} = 1.44$; another set of reproduced experimental results yielded nearly the same $K'_{eq,1-3}$ as can be seen in Fig. S9. The $K'_{eq,1-3}$ values were similarly estimated from electrochemical analysis in an aqueous acidic solution containing NaI as seen in Fig. S10, indicating that $K'_{eq,n}$ would not be or weakly related to a type of cations.

Lastly, $f_{I_{2n+1}^-(I_2-F)}$ was estimated from $K'_{eq,1-3}$ as can be seen in Fig. 5b. $f_{I^-(I_2-F)}$ fell to ~0 as $C_{I_2(I_2-F)}$ increased to 2.5 M, while $f_{I_3^-(I_2-F)}$ abruptly increased and became 0.4 at $C_{I_2(I_2-F)} = 0.73$ M. $f_{I_3^-(I_2-F)}$ decreased due to the formation of I_5^- and mostly I_7^- in I_2-F. I_7^- became the primary I_{2n+1}^- species in I_2-F at $C_{I_2(I_2-F)} \geqslant 5$ M.

Computational calculations for the stability of I_{2n+1}^- (n = 1–3) in water.—So far, we found that I_{2n+1}^- (n = 1–3) could exist in the vicinity of a Pt UME during the electro-oxidation of I^-, resulting in the formation of I_2-F. These species would transfer to a bulk aqueous solution; therefore, we should understand the relative stability of I_{2n+1}^- in water to better understand the mechanism for electro-oxidation of I^- in an aqueous solution, which was investigated through computational calculations.

All calculations were performed using the Gaussian 09 program[25] and visualized in Gaussview 6.[26] The geometric optimization of I_2, I_3^-, I_5^-, and I_7^- structures was performed using Couple Cluster theory including single and double excitations (CCSD) followed by single point calculations at Couple Cluster. Fig. 5. Experimentally obtained (a) $pI^-(I_2\text{-}F)$ vs of $C_{I^-(aq),free}$ and (b) $f_{I_{2n+1}^-(I_2-F)}$ vs $C_{I_2(I_2-F)}$ based on the mechanistic model for electro-oxidation of I^- as shown in Fig. 2. Single-Double excitations with perturbative triple excitation (CCSD(T)) theory levels with regard to the CCSD optimized geometries[27]; CCSD(T) is often referred to as the *gold standard* method for its accuracy and was demonstrated to be suitable for calculations of the ground-state energies of systems with a single reference character.[28] The effective core potential LanL2DZ (Los Alamos National Laboratory 2 Double-Zeta) basis set was used to describe the outermost valence electrons rather than considering time consuming full electron calculations.[29] The change in free energy of reaction 11 was then computed as the difference in free energy for the dissociation reaction occurring in the gas phase with that obtained in water solvent, employing the continuum solvation model based on the quantum mechanical charge density of a solute molecule interacting with a continuum description of the solvent (SMD) using default settings in Gaussian 09.[30] Vibrational analysis was performed at the same CCSD(T)/LanL2DZ theoretical level to ensure that all vibrational frequencies for each ground-state geometry were real.

The equilibrium geometries of I_2 and I_{2n+1}^- (n = 1, 2, and 3) in water computed at CCSD/LanL2DZ can be seen in Fig. 6. Thermodynamic properties such as the total electronic energy, including the zero-vibrational energy (ΔE in kcal mol^{-1}), free energy (ΔG in kcal mol^{-1}), enthalpy (ΔH in kcal mol^{-1}), and entropy (ΔS in cal mol^{-1} K^{-1}) of the inversed reaction 12 computed at CCSD(T)/lanL2DZ can be seen in Table I.

The geometric parameters of I_2^- and I_{2n+1}^- systems given in Fig. 6 approached the experimental geometries measured via X-ray based methods suggested elsewhere.[7,31] Deplano et al.[32] described I_7^- as a sequence of I_2-I_3^--I_2, which was similar to our computed I_7^- characterized by two short I-I 2.975 Å bonds and two long I-I 3.123 Å bonds that could be attributed with respect to I_2 and I_3^- (Figs. 6a–6b). The deviation in bond distances between the experimental[32] and theoretical calculations could be attributed to environmental influences (solid vs liquid).

The thermodynamic characteristics of the I_{2n+1}^- complexes formed by the combination of I_2 and I^- given in Table I computed at CCSD(T)/LanL2DZ showed that the formation of I_3^- in water occurred spontaneously ($\Delta G = -7.30$ kcal mol^{-1}), whereas the formation of I_5^-- was almost neutral ($\Delta G = 0.10$ kcal mol^{-1}) and that of I_7^- appeared to be endergonic ($\Delta G = 10.50$ kcal mol^{-1}). This indicated that I_5^- and I_7^- in water solvent tended to decompose to I^- and I_2, eventually combining to form I_3^-. Our results were in good

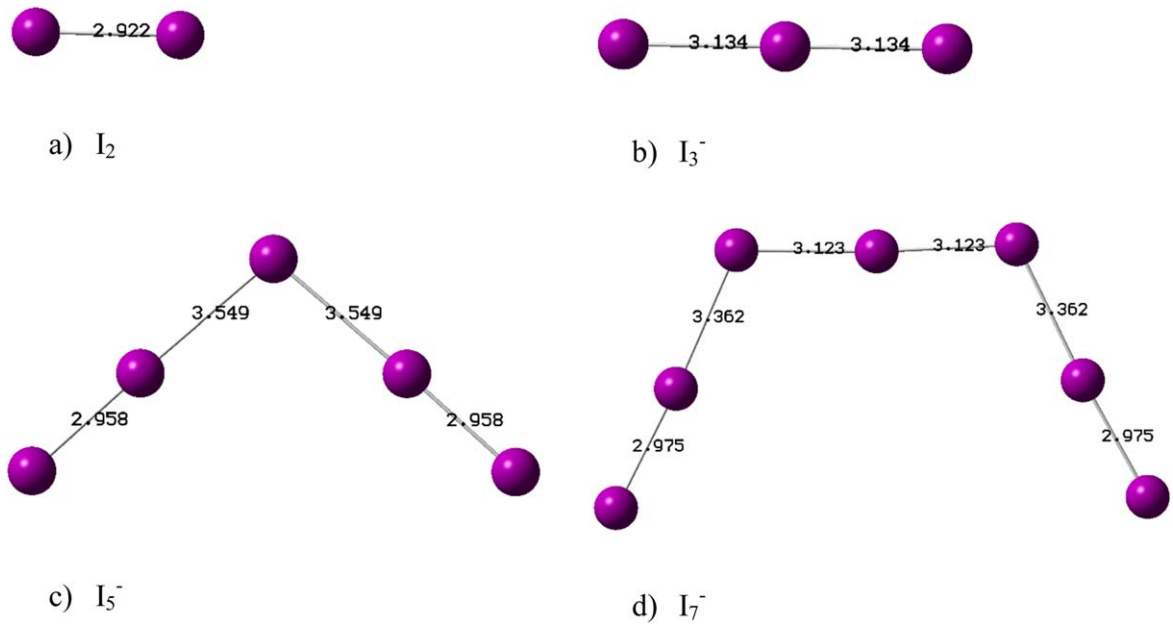

Figure 6. CCSD/LanL2DZ optimized geometries of (a) I_2, (b) I_3^-, (c) I_5^-, and (d) I_7^- in water; the unit of the all designated bond distances is Å.

Table I. Total electronic energy, including the zero-vibrational energy (ΔE in kcal mol^{-1}), enthalpy (ΔH in kcal mol^{-1}), free energy (ΔG in kcal mol^{-1}), and entropy (ΔS in cal mol^{-1} K^{-1}) of I_{2n+1}^- in water solvent.

I_{2n+1}^-	ΔE	ΔH	ΔG	ΔS	TΔS
I_3^-	−0.83	−0.01	−7.30	24.42	7.28
I_5^-	−2.38	−1.02	0.10	−3.76	−1.12
I_7^-	0.17	1.80	10.50	−29.16	−8.70

agreement with the report by Calabrese et al.[8] that I_3^- was the only isolated I_{2n+1}^- species that existed in an aqueous solution although they did not provide an explanation for the instability of larger polyiodides in aqueous solutions. ΔS for the formation of I_{2n+1}^--appeared to decrease with an increase in molecular size as seen in Table I. For instance, ΔS for the formation reaction of I_3^-- was nearly 24.42 kcal mol^{-1} whereas that of I_5^- and I_7^- was −3.76 and −29.16 kcal mol^{-1}, respectively. This indicated that both I_5^- and I_7^- were more structured than the corresponding reactants (I^- and nI_2) and I_3^-. The $T\Delta S$ energy (product of temperature and entropy) tended to cancel with ΔH in I_5^-, contrary to I_3^- where the $T\Delta S$ values were larger than ΔH. Those reactions were controlled by entropy (i.e. entropy-controlled reactions). Another factor that could justify the decomposition of I_{2n+1}^- to some extent into I_2 and I^- which rendered the formation of I_5^- and I_7^- non-spontaneous was the solvent effect. To understand solvent effects on I_{2n+1}^-, the solvation energy values of I^-, I_2, I_3^-, I_5^-, and I_7^- in water were evaluated at CCSD(T)/LanL2DZ, which can be seen in Table II.

Table II. Solvation energies of I^-, I_2, and I_{2n+1}^- (with n = 1, 2, 3) in water solvent in kcal mol^{-1}.

I_n^-	ΔE_{solv}	ΔH_{solv}	ΔG_{solv}
I^-	−63.98	−63.98	−63.98
I_2	−5.47	−5.48	−5.48
I_3^-	−37.74	−37.87	−37.87
I_5^-	−32.46	−32.45	−32.45
I_7^-	−29.02	−28.25	−28.25

The free energies of solvation (ΔG_{solv}) suggested I_2 (ΔG_{solv} = −5.48 kcal mol^{-1}), I_5^- (ΔG_{solv} = −32.45 kcal mol^{-1}), and I_7^- (ΔG_{solv} = −28.25 kcal mol^{-1}) to be less stable in water than I^- (ΔG_{solv} = −63.98 kcal mol^{-1}) and I_3^- (ΔG_{solv} = −37.87 kcal mol^{-1}). This change in thermodynamic parameters also supported the idea that I_{2n+1}^- should spontaneously move to I_3^-.

$f_{I_{2n+1}^-(I_2-F)}$ as a function of the overall kinetics of I_3^-(aq)-formation at various $C_{I^-(aq),free}$.—From the voltammetric titration and the computational calculations, we can consider that $C_{I_2(I_2-F)}$ would be governed by the rate of I^- electro-oxidation ($2I^- \rightarrow I_2 + 2e^-$) under a mass transfer controlled condition and that of I_3^--formation ($I^- + I_2 \rightarrow I_3^-$) with the corresponding overall rate constant, k_{f,I_3^-}, and $f_{I_{2n+1}^-(I_2-F)}$ could eventually be controlled by both k_{f,I_3^-} and $C_{I^-(aq),free}$ at equilibrium, which were estimated by finite element analysis.

The simulation domain with 2D axial symmetry, reactions, and the corresponding parameters can be seen in Fig. S11 and Table SII, respectively. We assumed a thickness of I_2-F at the steady state of I^- electro-oxidation is thin enough to consider that $C_{I_2(I_2-F)}$ is close to the surface concentration of I_2 at an electrode. Fig. S12 showed the plot of $C_{I_2(I_2-F)}$ vs $C_{I^-(aq),free}$ at various k_{f,I_3^-}. $C_{I_2(I_2-F)}$ to be linearly proportional to $C_{I^-(aq),free}$ at all k_{f,I_3^-}, and was extremely high without I_2-F dissolution reactions. However, at k_{f,I_3^-} = 1 M^{-1} s^{-1}, $C_{I_2(I_2-F)}$ significantly decreased and became reduced as k_{f,I_3^-} increased. In the simulation, we only consider diffusion because $C_{I_2(I_2-F)}$ as a function of $C_{I^-(aq),free}$ did not change even when migration was also considered as one of the mass transfers shown in Fig. S13; details of the corresponding time-dependent Nernst-Plank equations and parameters for the simulation with migration are listed in Table SIII.

Based on the estimated $C_{I_2(I_2-F)}$ as a function of $C_{I^-(aq),free}$ with different k_{f,I_3^-}, the contour plots of I_{2n+1}^- (n = 0–3) in I_2-F during the steady state associated with the electro-oxidation of I^- on a Pt UME were estimated as a function of k_{f,I_3^-} and $C_{I^-(aq),free}$ in Fig. 7. As shown in the fractional diagrams in Fig. 7a, it is difficult to find $I^-(I_2$-$F)$ in most of the plotted (k_{f,I_3^-}, $C_{I^-(aq),free}$) coordinates, indicating that $I^-(I_2$-$F)$ is highly unstable and reacts with I_2 to form I_{2n+1}^- in I_2-F. Both $I_3^-(I_2$-$F)$ and $I_5^-(I_2$-$F)$ primarily exist in the condition with $C_{I^-(aq),free} \leqslant 2$ M and $k_{f,I_3^-(aq)} \geqslant 100$ M^{-1} s^{-1}. $I_7^-(I_2$-$F)$ is the main polyiodide species at most (k_{f,I_3^-}, $C_{I^-(aq),free}$)

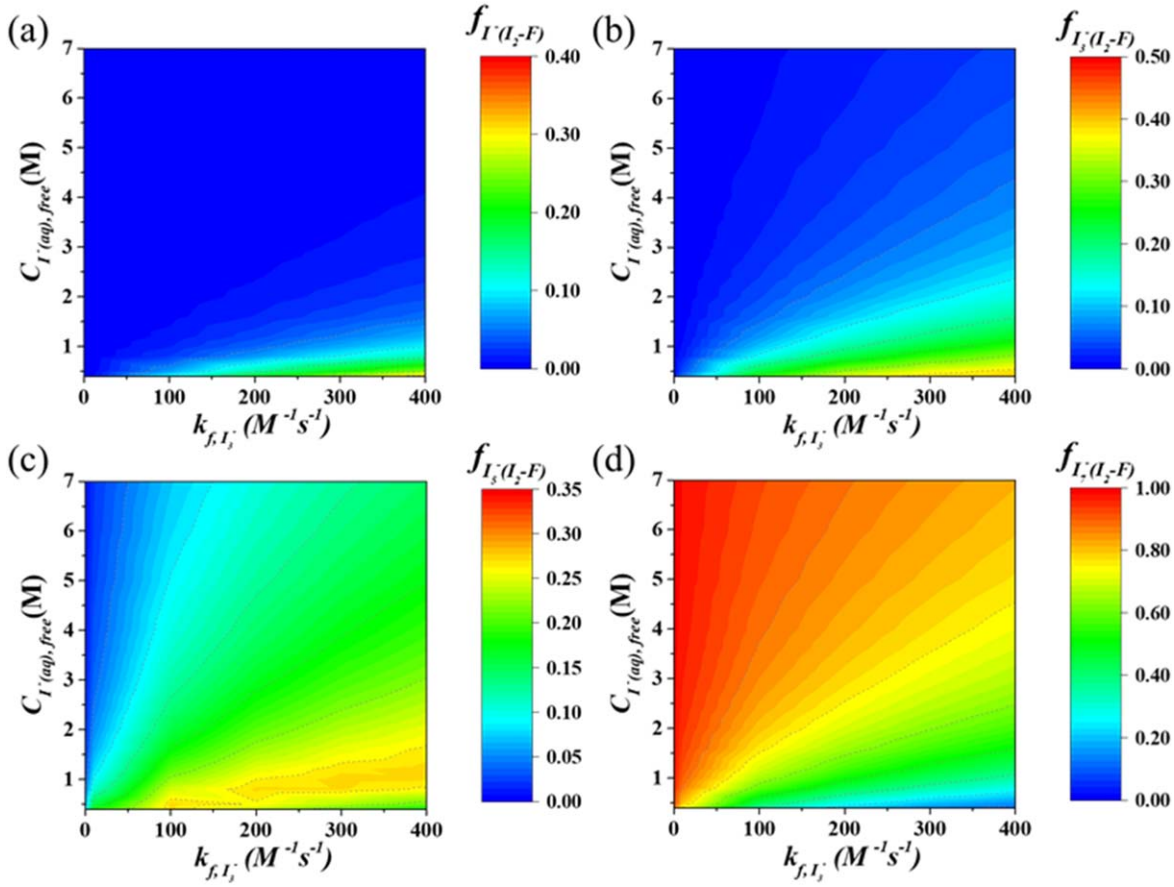

Figure 7. The contour plots of (a) $f_{I^-(I_2-F)}$, (b) $f_{I_3^-(I_2-F)}$, (c) $f_{I_5^-(I_2-F)}$, and (d) $f_{I_7^-(I_2-F)}$ as a function of both k_{f,I_3^-} and $C_{I^-(aq),free}$.

coordinates except for the region at $C_{I^-(aq),free} \leqslant 2.5$ M and $k_{f,I_3^-} \geqslant 39$ M^{-1} s^{-1}.

Summary and conclusive suggested mechanism for electro-oxidation of I^-(aq) to I_3^-(aq) via high orders of I_{2n+1}^- (n = 1–3).— In summary, we found electrochemically generated I_{2n+1}^- locally existing on a Pt UME during electro-oxidation of I^- in an acidic aqueous solution with highly concentrated ZnI_2 via voltammetric titration. I^- was electrochemically oxidized to form an iodine film (I_2-F). Then, in the model based on the proposed mechanism by Gileadi, I^- would be transferred fast through I_2-F by the Grotthus-type hopping mechanism. From the voltammetric titration analyses, we revealed I_7^- would be the primary species among I_{2n+1}^- (n = 0–3) in I_2-F, while I_3^- would be the main polyiodide in a bulk aqueous solution from the computational calculations, indicating that either I_5^- or I_7^- would only exist in the vicinity of a Pt UME during the electro-oxidation of I^-. The suggested overall reaction pathway for electro-oxidation of $I^-(aq)$ to $I_3^-(aq)$ via I_2-F is schematically described in Fig. 8. Lastly, we showed the local speciation of I_{2n+1}^- would depend on the overall kinetics of $I_3^-(aq)$ formation.

In many previous cases, electro-oxidation of I^- to I_2 and proceeding chemical I_3^--formation have been simply considered for the determination of the thermodynamic and the kinetic properties of the overall I^- electro-oxidation to I_3^- in an aqueous solution. From this study, however, we learned that electro-oxidation of I^- to I_3^- in an aqueous solution is, in fact, more complicated due to the existence of the complex polyiodides formations in I_2-F. Thus, physicochemical characteristics of polyiodides (I_3^-, I_5^-, and I_7^-)-formation in an iodine film should be carefully considered and studied to more deeply understand the I^-/I_3^- redox reaction via I_2-F in an aqueous solution, which would eventually give fundamental guidelines to develop Zn-polyiodide RFB.

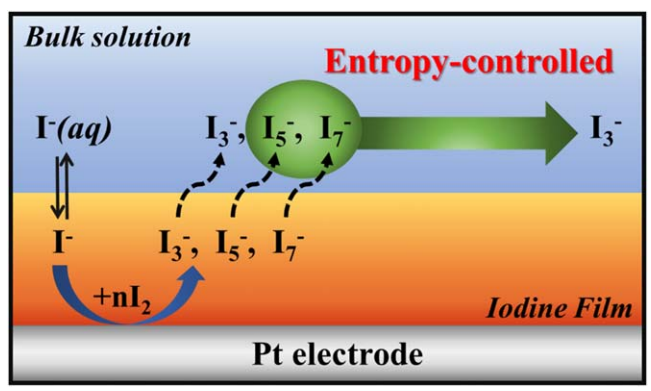

Figure 8. The schematic description for electro-oxidation of $I^-(aq)$ to $I_3^-(aq)$ via I_2-F.

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